Quarter1_Daily_Schedule

Quarter 1 — Day-by-Day Schedule (Foundations) · Sep 1 → early Dec 2026

Covers: the first ~13 weeks — **Week 0 (bio-clock + new-project ramp) → Phase 0 (fundamentals) → Phase 1 (math + NNs from scratch)** — ending Phase-1-complete and Tier-0-applying. Built on your blocks: **Block A 06:00–08:00 (peak: hard new theory)** · **Block B 08:30–10:30 (build: code/DSA)** · **Evening 20:30–22:30 (review/intake)** · **Weekend ~10 h buffer** (revision/spillover/emergencies). Target ~30 h/weekday-week; expect real delivery ~26–30 h — the weekend absorbs the gap.

> **Governing rules:** protect ~7 h sleep (lights-out ~22:45, wake ~05:45); the Evening block flexes first when tired (push to weekend); **self-tests are the gates** — if one isn't passed, repeat, don't advance. Consolidation weeks (5 & 12) are **no-new-material**. Threads T/M/R switch on only at the Phase-1 self-test (Week 10).

The default weekday template (fill with each week's content)

- **06:00–08:00 · Block A** — the single hardest *new* concept of the day, intuition-first: watch the visual explainer (≤30 min) → open the rigorous source → **derive/prove it on paper.** End by writing a 3-sentence glossary entry.
- **08:30–10:30 · Block B** — turn Block-A theory into code / solve problems. (From Week 10: first ~45 min = a thread, then phase coding.)
- **20:30–22:30 · Evening** — re-watch/lecture, read, **clear the Anki queue + glossary review**, re-derive today's Block-A concept once from memory, and note tomorrow's Block-A topic.
- **Daily spacing loop:** learn (A) → build (B) → review (Evening) → re-derive next morning.

WEEK 0 · Sep 1–6 — Ramp (bio-clock + project prep; ease in, ~15–20 h)

Goal: land the schedule and set up the environment; don't start hard learning yet.

- **Sleep shift:** each day move wake/sleep ~30 min earlier until you hit 05:45/22:45 by Sep 6.
- **Environment (one evening):** install Anki + create a "Confusion Buffer" deck; create a living-glossary doc; set up Python/conda + a GitHub repo `learning-log`; bookmark StatQuest, 3B1B, Karpathy Z2H.
- **New project (office hours):** scope the critic agentic doc-review pipeline; list its components (retrieve → draft-review → critique/verify → revise loop; where an MCP tool fits). *(This is office work, not a learning block — but note it maps to Phase 7.)*
- **Evenings (light):** watch 3B1B "Essence of Linear Algebra" ch 1–2 and one StatQuest overview — pure orientation, no notes pressure.
- **Sat/Sun:** rest + a 30-min read of the v12 plan so the arc is clear.

PHASE 0 — Fundamentals (Weeks 1–2) · ~60 h

Primary: StatQuest. Build the glossary aggressively. End with the articulation self-test → start applying.

Week 1 · Sep 7–13 — Statistics + classic-ML start (day-by-day)

Day	Block A (06:00–08:00)	Block B (08:30–10:30)	Evening (20:30–22:30)
Mon	Distributions, mean/variance/SD, populations vs samples (StatQuest)	In Python: simulate distributions; CLT demo by sampling	Re-watch CLT; Anki: 6 stat cards; glossary
Tue	Sampling, hypothesis testing, p-values	Code a permutation test + bootstrap CI	Re-watch p-values; Anki; glossary
Wed	Confidence intervals, MLE (intuition + one derivation)	Derive & code MLE for a Gaussian (mean/var)	Anki review (yesterday+today); glossary
Thu	Bias–variance; train/val/test & cross-validation	Code a bias-variance curve (poly-fit over-/under-fit)	Re-watch bias-variance; Anki
Fri	Confusion matrix; precision/recall; ROC & AUC vs PR	Code ROC/PR curves on an imbalanced toy set	Anki; write the week's 1-page "teach-a-junior" note
Sat/Sun	**Buffer:** finish any spillover; light review; rest one full day		

Week 2 · Sep 14–20 — Classic ML + DL fundamentals + self-test

Day	Block A	Block B	Evening
Mon	Linear & logistic regression; **why L1→sparsity (gradient-level)**	Code logistic regression + L1/L2 from scratch; watch weights zero out	Anki: regularization cards; glossary
Tue	Trees → RF → boosting → **GBM vs RF**	Fit XGBoost vs RF on a toy set; compare	Re-watch GBM; Anki
Wed	SVMs + the kernel trick; k-means; PCA (as variance/SVD preview)	Code k-means; PCA on a toy set	Anki; glossary

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Thu	Entropy, cross-entropy, mutual information; naive Bayes	Code entropy/CE; a tiny naive-Bayes classifier	Anki; glossary
Fri	DL fundamentals: neurons, layers, gradient descent (StatQuest NN + 3B1B NN ch1)	Code a 1-neuron gradient-descent loop by hand	**Assemble the articulation cheat-sheet (0.4)**
Sat	**PHASE 0 SELF-TEST (Feynman gate):** explain 15 concepts out loud, no notes — p-value, CI, CLT, MLE, bias-variance, ROC/AUC vs PR, L1-sparsity, GBM vs RF, entropy, gradient descent, etc.		
Sun	**▶ START APPLYING (Tier 0):** refresh CV/LinkedIn; send 3–5 quality applications (Qure.ai, SigTuple, GE, NVIDIA-India, MSR India)		

Gate: don't start Phase 1 until the self-test is clean. If shaky, use Week-3 mornings to shore up (and let Phase 1 shift a few days).

PHASE 1 — Mathematics + NNs from first principles (Weeks 3–12) · ~230 h

Block A = derivations (3B1B → MIT 18.06/Strang → derive by hand); Block B = implement in code + (from Wk 10) threads; Evening = re-watch + Anki + re-derive. This is the bedrock — do NOT rush.

Week 3 · Sep 21–27 — Linear algebra I

- **A:** 3B1B LA (vectors, linear combinations, span, linear transformations, matrix multiplication as composition) → begin MIT 18.06 (Strang) lectures 1–4.
- **B:** implement vectors/matrices, transformations, and matmul from scratch in NumPy; visualize transformations.
- **Evening:** re-watch the day's 3B1B; Anki (LA terms); glossary.
- **Daily A topics:** Mon determinants; Tue inverse & rank; Wed column/null space; Thu dot product & projections; Fri change of basis.

Week 4 · Sep 28–Oct 4 — Linear algebra II (eigen/SVD — the crux)

- **A:** eigenvalues/eigenvectors → **eigendecomposition → SVD** (3B1B eigen + MIT 18.06 + 18.065 SVD lecture). Derive SVD from eigendecomposition of A^TA on paper.
- **B:** code eigendecomposition & SVD; low-rank image compression with truncated SVD; verify A^TA is PSD.
- **Evening:** Anki (SVD/eigen — triangulate if slippery); glossary.
- **Whiteboard (Fri):** rank-r approximation (Eckart–Young); why A^TA is PSD.

Week 5 · Oct 5–11 — ◍ CONSOLIDATION (no new material, ~18 h)

- **A:** re-derive from memory: matmul-as-composition, determinant meaning, SVD-from-eigendecomposition, projection formula.
- **B:** re-implement (blank file): SVD-based PCA; a projection onto a subspace.
- **Evening:** full Anki catch-up; glossary cleanup; the week's teach-a-junior note on SVD.
- **Weekend:** rest; optional buffer for any lingering LA gap.

Week 6 · Oct 12–18 — LA finish + Calculus/matrix-calculus I

- **A:** finish 18.06 essentials (positive-definite, quadratic forms) → 3B1B Calculus (derivatives, chain rule) → **matrix calculus** (Parr & Howard).
- **B:** derive & code the gradient of a linear layer ($dL/dW = \delta \cdot a^T$); numerically check gradients.
- **Evening:** Anki; glossary; re-derive chain rule.

Week 7 · Oct 19–25 — Matrix calculus II + Probability I

- **A:** Jacobians/Hessians; gradient of softmax+cross-entropy (derive) → begin Stat 110 (Blitzstein): probability axioms, conditional probability, Bayes.
- **B:** code softmax+CE forward/backward from scratch; a Bayes-rule worked example in code.
- **Evening:** Anki; glossary; re-derive softmax-CE gradient.

Week 8 · Oct 26–Nov 1 — Probability II

- **A:** random variables, expectation/variance, common distributions, **conditional expectation as projection**, MLE revisited (Stat 110 / Blitzstein & Hwang).
- **B:** simulate distributions; MLE fitting; a small Bayesian update in code.
- **Evening:** Anki; glossary; whiteboard: $E[X|Y]$ as projection.

Week 9 · Nov 2–8 — Probability finish + NN-from-scratch I (micrograd)

- **A:** finish key probability (CLT proof sketch, covariance) → **Karpathy Z2H: micrograd** (autodiff, the chain rule as backprop).
- **B: build micrograd from scratch** (a scalar autodiff engine + a tiny MLP) — no looking at his code until stuck >30 min.
- **Evening:** re-watch micrograd; Anki (autodiff/backprop); glossary.

Week 10 · Nov 9–15 — NN-from-scratch II (makemore, backprop, attention) · Threads begin

- **A:** Karpathy Z2H: backprop mechanics, makemore (MLP language model), BatchNorm; then the attention video (scaled dot-product + $\sqrt{d_k}$).
- **B (threads on):** first 45 min = **Thread M** (reimplement: backprop through an MLP; then attention) / **Thread T** (start NeetCode arrays/two-pointer); rest = continue the Z2H build.
- **Evening:** re-derive backprop + $\sqrt{d_k}$; Anki; glossary; **Thread R** (skim one paper).
- **Fri: PHASE 1 SELF-TEST (gate):** hand-derive backprop for a 2-layer MLP; implement micrograd from memory; explain why BatchNorm helps + what breaks without it; whiteboard scaled dot-product attention and derive $\sqrt{d_k}$.

Week 11 · Nov 16–22 — Z2H finish (GPT) + Phase 2 kickoff · threads running

- **A:** Karpathy Z2H: build a small GPT (attention blocks, positional encodings) → **begin Phase 2.1 imaging physics** (MRI Q&A; fluorescence primer).
- **B:** finish the GPT build; **Thread M** (a sampler: top-k/temperature) + **Thread T** (binary search / sliding window).
- **Evening:** imaging-physics reading; Anki; **Thread R** (start a reproduce-a-paper pick).

Week 12 · Nov 23–29 — ◍ CONSOLIDATION (no new material, ~18 h + light threads)

- **A:** re-derive from a blank page: backprop, attention+$\sqrt{d_k}$, softmax-CE gradient, SVD.
- **B:** re-implement (blank file): micrograd MLP; multi-head attention.
- **Evening:** Anki catch-up; glossary cleanup; teach-a-junior note on attention.
- **Weekend:** rest; buffer for any Phase-1 gap.

Week 13 · Nov 30–Dec 6 — Bridge into Phase 2 (imaging)

- **A:** CNNs + convolution math (CS231n) — kernels, stride/padding, receptive fields; U-Net architecture.
- **B: Thread M** (implement a Dice loss + a conv2d from scratch) + **Thread T** (stacks/queues); set up the Capstone-1 dataset choice.
- **Evening:** U-Net / nnU-Net papers; Anki; Thread R.
- **End of Q1:** Phase 1 complete + threads live + Phase 2 underway. **Next quarter (Q2):** Phase 2 imaging core + expansion + Capstone 1 → Tier 0/1.

Weekend buffer — how to use the ~10 h

1. **Spillover first** — finish any weekday block you missed (life happens; this is expected).
2. **Then revision** — extra Anki, re-derive the week's hardest concept, re-implement one thing from a blank file.
3. **Deep-work option** — if caught up, merge into one ~4 h session on the current capstone/build (better continuity than weekday 2-h blocks).
4. **Protect one full day off.** If a week was rough, take *both* weekend days off and let the timeline slip — sustainable beats heroic.

Applications sub-track (from Week 2, ~1–2 h/wk in Evenings/weekend)

Week 2: CV/LinkedIn refresh + first Tier-0 applications. Weeks 3–13: ~3–5 quality applications/week; track in a sheet; treat early loops as paid calibration; add Capstone 1 to your profile as it lands (Q2).

The two Q1 gates (don't advance until clean)

- **Phase 0 (Week 2):** explain 15 fundamentals out loud, no notes.
- **Phase 1 (Week 10):** hand-derive backprop; micrograd from memory; BatchNorm why/what-breaks; attention + $\sqrt{d_k}$ at the whiteboard.